

Calculating Normal Probabilities

Graduate Project for MATH 5021

Ruoshui Li

Saint Louis University

Fall 2025

Background and Motivation

In his 1995 paper, Bagby discusses how to compute standard normal probabilities *without using tables*.

For a standard normal random variable X and $a > 0$, define

$$P(a) = \mathbb{P}(0 < X < a) = \frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx.$$

This integral cannot be expressed in terms of elementary functions. At the time the paper was written, personal computers were not yet universal, so finding a **simple and accurate approximation** for $P(a)$ was very important.

From One Dimension to Two Dimensions

The **first step** in the paper is to turn a one-dimensional integral into a two-dimensional integral.

Start with

$$P(a) = \frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx.$$

Then square both sides:

$$P(a)^2 = \left(\frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx \right)^2 = \frac{1}{2\pi} \left(\int_0^a e^{-x^2/2} dx \right) \left(\int_0^a e^{-y^2/2} dy \right).$$

Why Does the Product Become a Double Integral?

In general,

$$\left(\int_0^a f(x) dx \right) \left(\int_0^a f(y) dy \right) = \int_0^a \int_0^a f(x)f(y) dy dx.$$

In our case, $f(t) = e^{-t^2/2}$, so

$$P(a)^2 = \frac{1}{2\pi} \int_0^a \int_0^a e^{-(x^2+y^2)/2} dy dx.$$

This is exactly the step Bagby uses to write

$$P(a)^2 = \frac{1}{2\pi} \iint_{[0,a] \times [0,a]} e^{-(x^2+y^2)/2} dx dy,$$

which prepares for a polar coordinate change and later for a new quadrature rule.

Why “Sum of Sum” Becomes a Double Sum

Consider two sums: Let

$$A = \sum_{i=1}^n a_i, \quad B = \sum_{j=1}^m b_j.$$

Then

$$A \cdot B = \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^m b_j \right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j.$$

A simple example: take $n = m = 2$,

$$\begin{aligned} \left(\sum_{i=1}^2 a_i \right) \left(\sum_{j=1}^2 b_j \right) &= (a_1 + a_2)(b_1 + b_2) \\ &= a_1 b_1 + a_1 b_2 + a_2 b_1 + a_2 b_2 = \sum_{i=1}^2 \sum_{j=1}^2 a_i b_j. \end{aligned}$$

From Double Sum to Double Integral

For integrals, the Riemann sums look like

$$\sum_i f(x_i) \Delta x, \quad \sum_j f(y_j) \Delta y,$$

and their product becomes

$$\sum_i \sum_j f(x_i) f(y_j) \Delta x \Delta y.$$

Changing to Polar Coordinates

From the previous steps, Bagby shows that

$$P(a)^2 = \frac{1}{\pi} \int_0^{\pi/4} \int_0^{a \sec \theta} e^{-r^2/2} r \, dr \, d\theta.$$

Here we have already switched to **polar coordinates**

$$x = r \cos \theta, \quad y = r \sin \theta,$$

so that $x^2 + y^2 = r^2$ and $dx \, dy = r \, dr \, d\theta$.

The limits $0 \leq \theta \leq \pi/4$ and $0 \leq r \leq a \sec \theta$ describe the region in the first quadrant under the line $y = x$ and inside the circle $r = a \sec \theta$.

From Double Integral to Bagby's Formula

Starting from

$$P(a)^2 = \frac{1}{\pi} \int_0^{\pi/4} \int_0^{a \sec \theta} e^{-r^2/2} r \, dr \, d\theta,$$

we first integrate with respect to r (note that the upper limit depends on θ):

$$\int_0^{a \sec \theta} e^{-r^2/2} r \, dr = \left[-e^{-r^2/2} \right]_0^{a \sec \theta} = 1 - e^{-\frac{1}{2}a^2 \sec^2 \theta}.$$

Thus

$$P(a)^2 = \frac{1}{\pi} \int_0^{\pi/4} (1 - e^{-\frac{1}{2}a^2 \sec^2 \theta}) \, d\theta = \frac{1}{\pi} \int_0^{\pi/4} 1 \, d\theta - \frac{1}{\pi} \int_0^{\pi/4} e^{-\frac{1}{2}a^2 \sec^2 \theta} \, d\theta.$$

Since $\int_0^{\pi/4} 1 \, d\theta = \frac{\pi}{4}$, we obtain

$$P(a)^2 = \frac{1}{4} - \frac{1}{\pi} \int_0^{\pi/4} e^{-\frac{1}{2}a^2 \sec^2 \theta} \, d\theta.$$